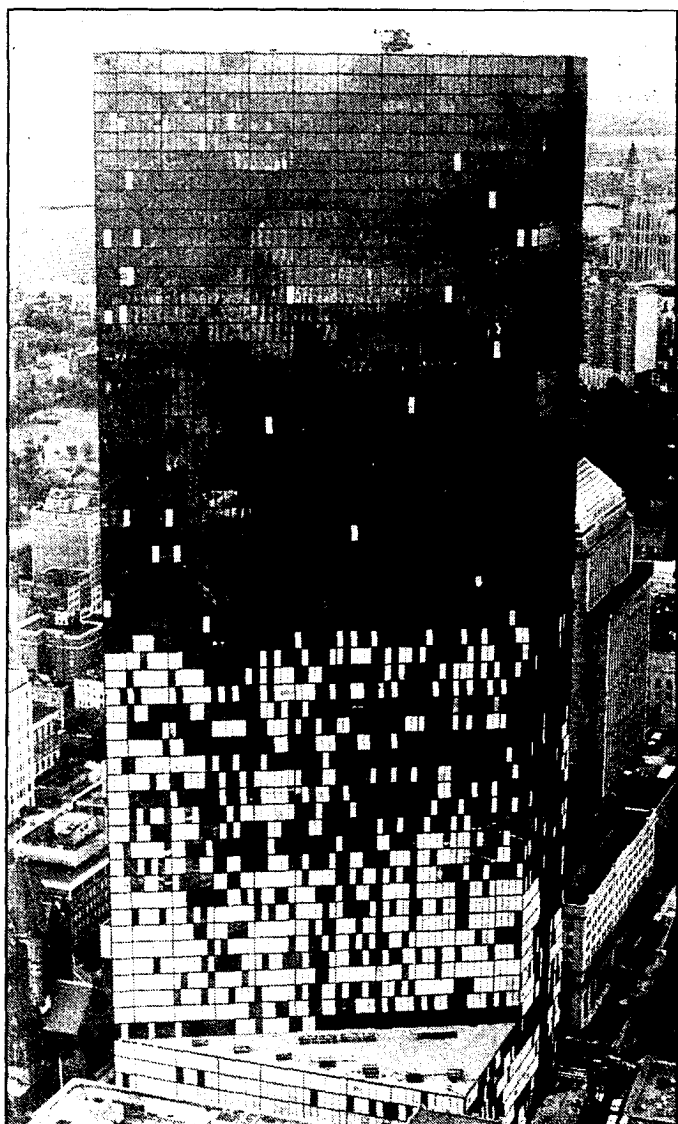


IT WAS A TOWER OF TRIBULATION



Globe file photo

The Hancock Tower was speckled with plywood in 1973 as engineers tried to figure out why its windows were cracking.

How a Swiss engineer may have saved the Hancock building from toppling

By Robert Campbell
Special to the Globe

Few Bostonians — few Boston architects, in fact — realize that there was a time when the Hancock Tower was considered to be in danger of falling over on its edge.

The risk that the tower might collapse was only one of a series of technical problems

that haunted the building during the years when it was under construction.

Today, the Hancock is one of Boston's most prominent, most admired architectural landmarks. But 15 years ago, it was the unfortunate protagonist of an architectural sequel to "The Perils of Pauline."

What really happened to the Hancock? The full story has never come out, because of a nondisclosure agreement signed by the parties involved in litigation over its construction — the owner, architect, general contractor, glass man-

ufacturer and others. The agreement forbids any of them — "in perpetuity" — from discussing the Hancock's problems.

This article is based on interviews with two persons who were in a position to observe the Hancock closely during its time of troubles but who were not directly involved in the litigation. They are William LeMessurier, a Cambridge structural engineer, and Victor Mahler, a consultant on curtain-wall technology in New York.

This, then, is what happened to the Hancock.

The building is a 60-story, 790-foot, mirror-glass office tower designed by Henry N. Cobb of I. M. Pei & Partners. Its skeleton is a conventional steel frame. Its skin consists of 10,344 identical glass panels, each about 4½ by 11½ feet. Its physical form is unusual in one respect: it is very narrow in one

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direction and very wide in the other.

The Hancock's problems surfaced in a dramatic manner. On the night of Jan. 20, 1973, while the tower was under construction, a high wind blew in Boston. By morning, dozens of the building's glass panes had failed, some shattering and falling to the pavement. More windows cracked in subsequent weeks.

Eventually, about a third of the glass was removed for safety's sake and replaced with plywood. Jokes were made about the world's tallest wooden building. Finally, the entire tower was re-clad with a different type of glass.

There were other problems. Buildings nearby, such as the Copley Plaza Hotel and Trinity Church, began to settle and crack. Underground utility lines were damaged.

Then it was reported that the Hancock's steel frame was being stiffened with new steel. It was reported, too, that something called a tuned mass damper was being installed on an upper floor to keep the tower from swaying in the wind.

In 1976, the tower was finally finished and opened for business. Its spectacular, if rather aloof, beauty quickly began to erase the recollection of its technical problems. But there are lessons to be learned from the Hancock's troubles that deserve to be remembered.

Four different problems

To understand what happened to this building, you have to grasp one essential fact. It is that the Hancock actually suffered not one but four technical failures, and none of them had anything to do, in principle, with any of the others.

Failure No. 1 was that, when the hole for the Hancock's foundation was dug, its walls caved in.

Failure No. 2 was that the windows fell out.

Failure No. 3 was that the tower, when built, was found to be swaying too much in the wind.

Failure No. 4 — easily the most remarkable — was that the tower was found to be in danger of actually falling over.

Remember, none of these problems had anything to do with any of the others. For example, the windows didn't fall out because the building was swaying in the wind — though it was. They have to be taken one by one:

1. The cave-in of the foundation.

While the Hancock site was still a hole in the ground — an excavation for the future tower — the sides of that hole were insufficiently braced. They collapsed into the hole, in some cases as much as 3 feet. That partial cave-in caused earth all around the excavation to shift.

Because of the moving earth, Trinity Church, just across the street from the hole, suffered very severe problems of movement and settling. In one case, a wall of the church began to tilt outward almost to the point at which the roof would have collapsed.

This cave-in, which occurred before the tower was even built, is what caused all the broken utility lines and the settling of other buildings around the Hancock. Despite persistent rumors, the Hancock itself — built on steel piles that were sunk to bedrock — never settled at all.

2. The falling windows.

The Hancock windows, once they began to fall, became surely the most closely studied detail in the history of architecture. On-site investigations were mounted, laboratory mock-ups were constructed, and wind-tunnel tests of every kind were made.

At first, everyone assumed that the problem was caused either by the wind or by some kind of movement of the building, or both. Maybe the building was settling. Maybe its sharp angles were causing weird wind velocities. Maybe it

was simply swaying in the breeze and thus cracking its glass.

None of those perfectly logical conjectures turned out to be true. The problem, it was found, was entirely in the typical glass window itself. The Hancock window failure was a product failure.

The Hancock window, assembled in the factory, was double-glazed — it was a sandwich of two panes of glass with air between them, separated by a lead spacer strip. On the inside surface of the outside pane was a silver coating — then a new product — to reflect heat and glare away from the building. The lead spacer was soldered to both panes of glass, which meant that in the case of the outside pane it was soldered to the silver coating.

Joint wouldn't flex

The source of the falling windows proved to be this tiny soldered joint. The joint, apparently, was simply too good. It was so strong, so rigid, that it lacked flexibility.

As the window expanded and contracted normally with changes in temperature, and as it vibrated normally in the wind, the joint on the outside wouldn't flex. Microscopic cracks would open at the edge of the outside pane and would gradually enlarge until the window was so weakened it would fail.

As you might expect, it was eventually found that many other buildings using this same new type of double-glazed window saw them fail at about the same time the Hancock's did. But those buildings were much smaller and their failures much less spectacular.

When the Hancock was re-glazed, single panes of thick reflective glass were used. They proved successful from the first.

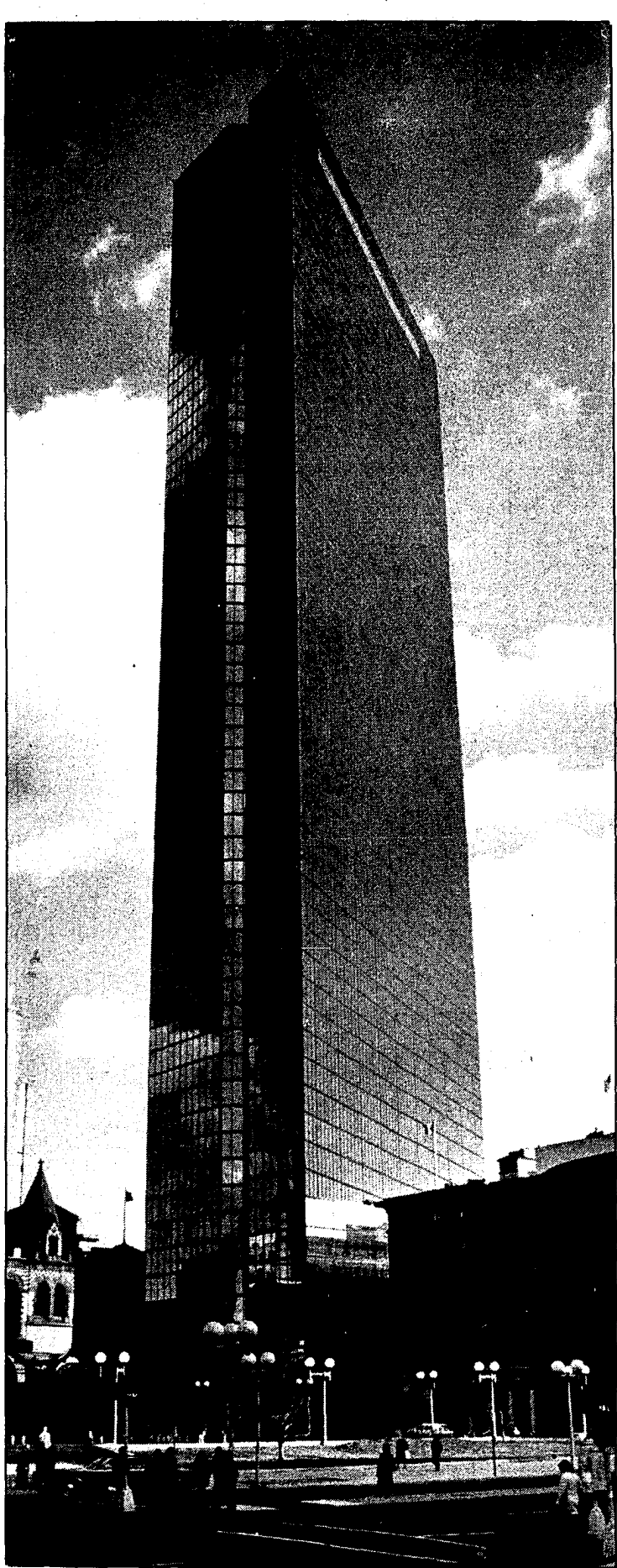
3. Swaying in the wind.

The window problem led to an unprecedented number of studies of the Hancock, and one of the things examined was how it moved in the wind. The studies showed that the tower might sway or twist in high winds too much for the comfort of its occupants. This was a problem that couldn't have been predicted, simply because the science of comfort in moving buildings was barely in its infancy at the time the Hancock was designed.

The problem was corrected by what engineers call a "tuned mass damper," a device invented by William LeMessurier and others for the Citicorp tower in New York. At the Hancock, the damper consists of two massive blocks of lead, each weighing about 300 tons. They are mounted at opposite ends of the 58th floor and are free to slide back and forth on steel plates covered with a film of oil. Attached to the building's steel frame by springs and shock absorbers, they act as dampers to reduce the amount by which the tower sways and twists in ordinary strong winds.

4. The danger of falling down. This last problem is the really scary one, and the most fascinating of the Hancock's tribulations. One reason it's so scary is that it never would have been discovered except for the concern over the windows.

As noted above, the window failure brought the Hancock under intense scrutiny by engineers. Some of them became concerned that the building might conceivably be in danger of collapse in rare, powerful windstorms. To understand what followed, it helps to think of the Hancock as a huge book, standing upright on its bottom edge as a book stands on a bookshelf. The engineers were thinking entirely of one kind of movement. They were worried that extraordinary winds might cause the tower to fall over the way a book might be blown over, on one of its flat sides. Nobody dreamed that the building could ever collapse *endwise*, any more than you'd dream a book could fall over on its thin edge.



Globe file photo
Window problems solved, the Hancock Tower was opened in 1976.

To explore the fears the engineers did have, the architect, Henry Cobb, decided to get a once-and-for-all answer from an internationally known expert on the behavior of tall buildings, a structural engineer named Bruno Thurlimann in Zurich, Switzerland. Working with a wind tunnel expert in Ontario, Thurlimann gave the Hancock probably the most complete analysis any tower had ever undergone. The result of his investigation has something of the drama of a detective story.

Thurlimann was looking, as everyone had all along, at the possibility that the building would collapse on its wide side. And he concluded that the Hancock was more than strong and stiff enough to resist the overturning force of the wind against those sides, and that it was therefore safe. He was about to fly to the United States to deliver this cheerful final report when he decided that, just as a matter of professional thoroughness, he ought to look at the tower's behavior along its long axis — in other words, at the possibility of its falling on its edge.

When he did, he made an astonishing discovery. The Hancock Tower was in danger, under certain quite possible wind conditions, of overturning that way.

Thurlimann's analysis was compelling enough to persuade the Hancock's architects, engineers and owners to take quick and drastic action. At a cost of several million dollars, the tower's steel frame was braced with new diagonal steel struts from top to bottom, doubling its stiffness along the long axis.

It wasn't only the Hancock that was stiffened. So was the national structural-steel building code — because the Hancock had met or exceeded that code in every respect. The code, it turned out, hadn't been dealing adequately with the behavior of towers like the Hancock that are very long in one direction. It was the Hancock's height and its great length along two sides — not its thinness and not its sharp angles — that made it vulnerable.

The Hancock survived its problems, as everyone knows, to win awards for its cool beauty and to become perhaps the most distinctive building on the Boston skyline. It left a legacy, too, of better engineering. Wind-tunnel methods, structural-steel analysis and glass-curtain-wall technology all improved because of the Hancock. Henry Cobb, whose performance during the long, agonizing crisis is regarded by fellow architects as a model of professional integrity, went on to become, for five years, chairman of the department of architecture at Harvard and the designer of many notable buildings.

Today, it's hard even to find photographs of the Hancock when it was the Plywood Tower.

But the lesson of the Hancock shouldn't be forgotten. It isn't the only major American building to have had a brush with disaster in recent years. There is an analogy with what happened to the space shuttle program. The lesson is that the ambitions of designers, builders and product manufacturers can sometimes outrun the available technology. The more we all know about such events, the better off we all are.